



NLP-Based Intelligent Food Recipe Recommendation and Diet Planning System

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

Capstone Project – Artificial Intelligence & Natural Language Processing

to the award of the degree of

**BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING**

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "**NLP-Based Intelligent Food Recipe Recommendation and Diet Planning System**" has been carried out by N. Goutham (192472024), under the supervision of Dr. Anbazhagan K and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer Science and Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, N. Goutham, of the Department of Computer Science and Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "NLP-Based Intelligent Food Recipe Recommendation and Diet Planning System" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: 03-09-2026

Signature of the Students with Names

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INDIVIDUAL CONTRIBUTION STATEMENT

Student Name / Reg. No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
N. Goutham 192472024	NLP preprocessing pipeline; ingredient extraction	Text cleaning, tokenization, lemmatization, TF-IDF vectorization module	Unit testing of NLP pipeline on sample inputs	Ch. 1, Module 1 (4.2)	20%
A. Shiva Shankar Reddy 192425174	Recipe retrieval & recommendation engine	Recipe API integration, Cosine Similarity ranking logic	Validation of similarity ranking accuracy	Ch. 2, Module 2 (4.2)	20%
B.V.N.S. Bhimeswar 192472048	Nutrition analysis & diet planning module	Rule-based dietary filtering engine, nutrition scoring	Testing of diet-goal based filtering rules	Ch. 3, Module 3 (4.2)	20%
M. Tarun Prasad 192425024	Recipe generation, multilingual output & UI/UX	Natural language generation for recipe steps, multilingual translation integration, front-end (Recipe Lab / Generation Gourmet / Body Architect screens)	End-to-end system testing, UI testing across languages	Ch. 4, Module 4 (4.2), consolidation of report	20%
N. Lokesh 192425327	User profile, history & favorites management	Session/user data management, history and favorites storage	Testing of persistence, save/retrieve flows	Ch. 5, Module 5 (4.2)	20%
Total					100%

ABSTRACT

Selecting recipes that match the ingredients a person already has, while also meeting personal dietary goals, is a recurring and often frustrating problem, since existing recipe platforms mostly rely on keyword search and provide limited personalization. This project addresses that engineering problem by designing and implementing an NLP-based AI Food Recipe Generator and Personalized Diet Planner that recommends suitable recipes from a user's available ingredients and generates a nutrition-aware diet plan aligned with the user's fitness goals.

The system is needed because users struggle to find recipes that use only the ingredients on hand, existing systems seldom combine ingredient-based retrieval with nutritional analysis, and there is a growing demand for multilingual, easy-to-understand cooking guidance for a diverse user base. The proposed solution accepts free-text ingredient input together with meal type, diet goal, age group and language preference, and returns ranked, personalized recipes and meal plans.

The methodology follows a five-module pipeline. Module 1 performs NLP preprocessing — text cleaning, tokenization and lemmatization — followed by TF-IDF vectorization of the ingredient text. Module 2 retrieves candidate recipes from an integrated recipe database/API and ranks them using Cosine Similarity between the user vector and each recipe vector. Module 3 analyses nutritional attributes (calories, protein, carbohydrates, fat, fibre) and applies rule-based filtering against the user's stated diet goal. Module 4 generates human-readable, multilingual recipe output using natural language generation and translation. Module 5 manages the user's profile, history and favourites so that recommendations improve with repeated use.

The system was implemented as a web application ('AI Recipe') with three coordinated experiences — Recipe Lab (ingredient-based recommendation), Generation Gourmet (age- and meal-time-based suggestions) and Body Architect (goal-driven 7-day meal planning). Testing against representative ingredient sets and diet goals showed that the TF-IDF and Cosine Similarity pipeline consistently retrieved recipes that matched the majority of user-supplied ingredients, that nutrition-based filtering correctly excluded recipes violating the stated diet goal, and that multilingual generation (demonstrated in English and Telugu) produced coherent, usable output.

The principal conclusion is that combining lightweight, interpretable NLP techniques (TF-IDF and Cosine Similarity) with rule-based nutrition filtering is sufficient to deliver personalized, explainable recipe and diet recommendations without the computational overhead of large language models, while still leaving a clear path — via the Feedback & Learning module — toward learning-based personalization in future iterations.

Keywords: *Natural Language Processing, TF-IDF, Cosine Similarity, Recipe Recommendation, Personalized Nutrition, Multilingual Generation*

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description
NLP	Natural Language Processing
TF-IDF	Term Frequency – Inverse Document Frequency
NLG	Natural Language Generation
API	Application Programming Interface
RAM	Random Access Memory
kcal	Kilocalories
UI/UX	User Interface / User Experience

CHAPTER 1 INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Recipe discovery has traditionally relied on keyword search across large recipe repositories, where the user searches by dish name or a single primary ingredient and manually filters through a long list of results. With the growth of Natural Language Processing (NLP) and information-retrieval techniques such as TF-IDF (Term Frequency–Inverse Document Frequency) and Cosine Similarity, it is possible to move beyond keyword search toward ingredient-driven, similarity-based recommendation, where a free-text list of ingredients supplied by the user is matched against a corpus of recipes. Combining this retrieval layer with rule-based nutritional filtering and Natural Language Generation (NLG) allows a system to produce not just a ranked list of recipes but a complete, readable, and even multilingual cooking and diet-planning experience.

1.2 Need for the Project

- Home cooks frequently have a fixed set of ingredients on hand and need recipes that make effective use of exactly those ingredients, rather than requiring a shopping trip.
- Individuals following specific diet goals (high protein, low calorie, weight management, muscle building) are affected by the lack of nutrition-aware recipe filtering in most consumer recipe platforms.
- Existing recipe platforms provide keyword search but rarely combine ingredient matching, nutrition analysis and personalized diet planning in a single system.
- Non-English-speaking users are underserved, since most mainstream recipe platforms provide content only in English.
- From an engineering standpoint, there is significant value in demonstrating that classical, interpretable NLP techniques (TF-IDF, Cosine Similarity, rule-based filtering) can deliver a usable personalization system without the cost and opacity of large language models.

1.3 Problem Statement

Given a free-text list of ingredients, a stated meal type, a diet goal, an age group and a preferred output language, design and implement a system that (i) linguistically processes the ingredient text, (ii) retrieves and ranks recipes that best match the available ingredients, (iii) filters and scores those recipes against the nutritional requirements implied by the diet goal, and (iv) generates a clear, structured, multilingual recipe and diet-plan output — while resolving the conflicting requirements of retrieval accuracy, nutritional correctness, response latency and interpretability of results.

1.4 Project Aim

To design, develop and evaluate an NLP-based AI Food Recipe Generator and Personalized Diet Planner that recommends recipes from user-supplied ingredients and produces personalized, multilingual, nutrition-aware meal recommendations.

1.5 Project Objectives

1. To design an NLP preprocessing pipeline (cleaning, tokenization, lemmatization) for user-entered ingredient text.
2. To develop a TF-IDF based vector representation for ingredients and recipes.
3. To model a Cosine-Similarity-based recipe retrieval and ranking engine.
4. To implement a rule-based nutrition analysis and diet-filtering module.
5. To test the system's retrieval accuracy, filtering correctness and multilingual output across representative user scenarios.
6. To evaluate the system against existing recipe-recommendation approaches identified in the literature review.

1.6 Scope

Included:

- Ingredient-based recipe retrieval and ranking using TF-IDF and Cosine Similarity.
- Nutrition analysis and rule-based diet-goal filtering (calories, protein, carbohydrates, fat, fibre).
- Multilingual recipe generation (demonstrated in English and Telugu).
- User profile, recipe history and favourites management.
- A 7-day meal-plan generator (Body Architect) based on fitness goal, body metrics and diet preference.

Excluded:

- Real-time grocery price comparison or grocery ordering integration.
- Computer-vision-based ingredient recognition from images.
- Clinical-grade dietary advice for medically diagnosed conditions.

Assumptions:

- The recipe database/API used provides sufficiently rich ingredient and nutrition metadata for each recipe.
- Users provide ingredient names in a supported language, using common naming conventions.

Boundary Conditions:

- Recommendation quality is bounded by the coverage and quality of the underlying recipe corpus.
- Translation quality for the multilingual module is bounded by the accuracy of the integrated translation service.

1.7 Expected Engineering Outcome

The expected outcome is a working, testable web-based software system (a recipe-recommendation and diet-planning application) comprising an NLP processing pipeline, a similarity-based retrieval and ranking model, a rule-based nutrition-filtering engine, a natural-language / multilingual generation layer, and a user-profile management layer — packaged as an integrated, demonstrable prototype.

CHAPTER 2 LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature was identified through a targeted search of recent (2023–2026) peer-reviewed conference and journal papers on recipe recommendation, food-ingredient recognition, and nutrition-oriented recommender systems, supplemented by a review of commercial recipe platforms to assess prevailing product-level capabilities and gaps.

2.2 Review of Existing Work

Five representative works were critically reviewed, spanning ingredient-recognition-based retrieval, hybrid collaborative/content-based recommendation, biometric health-aware recommendation, LLM-based conversational nutrition chatbots, and TF-IDF/Cosine-Similarity-based retrieval. Table 2.1 summarizes the reviewed literature.

Table 2.1: Literature Survey Summary

S.No	Author(s)	Year	Paper Title	Method / Focus	Limitations
1	M. A. et al.	2023	RecipeIS — Recipe Recommendation System Based on Recognition of Food Ingredients	Ingredient-based recipe recommendation	Limited focus on personalized nutrition planning
2	K. Zioutos et al.	2024	Healthy Personalized Recipe Recommendations for Weekly Meal Planning	Hybrid collaborative and content-based recommendation	Requires detailed user and health information
3	Y. Lao et al.	2024	Dynamic Personalized Recipe Recommendations Based on Facial Health Recognition	Health-aware personalized recipe recommendation	Depends on facial health recognition complexity
4	Z. Yang et al.	2024	ChatDiet: Personalized Nutrition-Oriented Food Recommender Chatbots	LLM-based conversational nutrition recommendation	Higher computational complexity and LLM dependency
5	J. Wong & Y. Yoannita	2026	TF-IDF and Cosine Similarity for Indonesian Recipe Recommendation	TF-IDF, preprocessing and Cosine Similarity	Limited advanced nutrition and multilingual features

2.3 Comparative Analysis

Table 2.2: Comparative Analysis of Existing Approaches

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Ingredient-recognition-based recommendation (RecipeIS)	Automates ingredient identification	Weak nutrition personalization	Motivates explicit nutrition module in proposed system

Hybrid collaborative + content-based (Zioutos et al.)	Balances personalization sources	Needs extensive user/health data upfront	Proposed system keeps input lightweight (ingredients + goal)
Facial-health-recognition-based (Lao et al.)	Novel health-aware signal	Complex, privacy-sensitive, hardware dependent	Proposed system avoids biometric dependency
LLM-based conversational (ChatDiet)	Flexible, conversational UX	High computational cost, less interpretable	Proposed system favours interpretable TF-IDF + rules
TF-IDF + Cosine Similarity (Wong & Yoannita)	Lightweight, interpretable, fast	Limited nutrition and multilingual features	Proposed system extends this approach with nutrition filtering and multilingual NLG

2.4 Research / Engineering Gap

- No reviewed system combines lightweight, interpretable ingredient-similarity retrieval with explicit rule-based nutrition filtering and multilingual generation in a single pipeline.
- Systems that add richer personalization (hybrid recommenders, biometric signals, LLM chatbots) do so at the cost of higher data requirements, computational cost, or reduced interpretability.
- Multilingual recipe output — important for a linguistically diverse user base — is largely unaddressed in the surveyed literature.

2.5 Proposed Contribution

The proposed system differs from existing approaches by combining (i) a lightweight NLP preprocessing and TF-IDF vectorization stage, (ii) Cosine-Similarity-based recipe retrieval and ranking, (iii) an explicit, interpretable rule-based nutrition and diet-goal filtering stage, and (iv) a natural-language-generation and multilingual translation stage — all within a single, low-latency pipeline that does not depend on large language models or biometric sensing, while still supporting personalized, multi-day meal planning through the Body Architect module.

CHAPTER 3 ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

Table 3.1 Stakeholder / User Requirements

Stakeholder	Requirement
End user (home cook)	Recommend recipes usable with the ingredients already on hand, in an understandable language
Health-conscious user	Recommend recipes consistent with a stated diet goal (e.g., high protein, low calorie)
Non-English-speaking user	Provide recipe names, ingredients and instructions in the user's preferred language
Project supervisor / evaluator	Demonstrate a working, testable pipeline with measurable retrieval and filtering accuracy

Table 3.2 Functional & Non-Functional Requirements

Requirement Type	Requirement	Measurable Target
Functional	Accept free-text ingredients and preprocess (clean, tokenize, lemmatize)	100% of test inputs successfully tokenized and lemmatized
Functional	Vectorize ingredients using TF-IDF	Vector generated for every valid input within the test set
Functional	Rank recipes using Cosine Similarity	Top-N (N=5) recipes returned per query
Functional	Filter recipes by nutrition/diet goal	0% of returned recipes violate the stated diet-goal rule
Functional	Generate multilingual recipe text	Output verified correct in ≥ 2 languages (English, Telugu)
Non-Functional	Response latency	Recommendation generated in < 3 seconds per query on target hardware
Non-Functional	Usability	Task (get a recommended recipe) completed by a first-time user in < 2 minutes

Table 3.3 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
ISO/IEC 25010 (Software Quality)	Functional suitability, performance efficiency, usability	Used as an informal checklist for module-level testing (Section 4.3)
Data protection / privacy	User data (profile, history) handled	User history and favourites stored per-

practice (institutional guideline)	responsibly	user with no third-party sharing
W3C Internationalization (i18n) guidelines	Consistent handling of multilingual text	Applied in the multilingual output module (Module 4)

Table 3.4 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Recommendation latency	< 3	seconds	High
Top-N recipes returned	5	recipes	Medium
Minimum ingredient match ratio for a recommended recipe	≥ 60	%	High
Supported output languages (demonstrated)	≥ 2	languages	Medium
Nutrition attributes analysed per recipe	5 (calories, protein, carbs, fat, fibre)	attributes	High

3.4 Alternative Solutions & Evaluation

Alternative 1 (Keyword Search): Recipes are matched by exact or partial keyword overlap with user-entered ingredient text; simple to implement but poor at handling ingredient synonyms or partial matches.

Alternative 2 (TF-IDF + Cosine Similarity): Ingredient text and recipe ingredient lists are vectorized using TF-IDF and ranked by cosine similarity, giving weighted, partial-match-tolerant retrieval.

Alternative 3 (LLM-based Recommendation): A large language model interprets ingredient text and directly proposes recipes; flexible but computationally expensive and less transparent.

Table 3.5 Alternative Solution Evaluation Matrix

Criterion	Weight	Alt. 1: Keyword Search	Alt. 2: TF-IDF + Cosine Similarity	Alt. 3: LLM-based Recommendation
Performance (latency)	30%	High	High	Low
Cost (compute)	20%	High	High	Low
Accuracy of ingredient match	25%	Low	High	High
Interpretability	15%	High	High	Low
Ease of extending with nutrition rules	10%	Medium	High	Medium

3.5 Selected Design & Detailed Design

Alternative 2 (TF-IDF + Cosine Similarity) was selected based on the evaluation matrix in Section 3.4: it scored highest on accuracy of ingredient matching and interpretability while retaining the low latency and low cost of the keyword-search baseline, avoiding the compute cost and opacity of the LLM-based alternative. This directly supports the non-functional latency and usability targets set in Section 3.3.

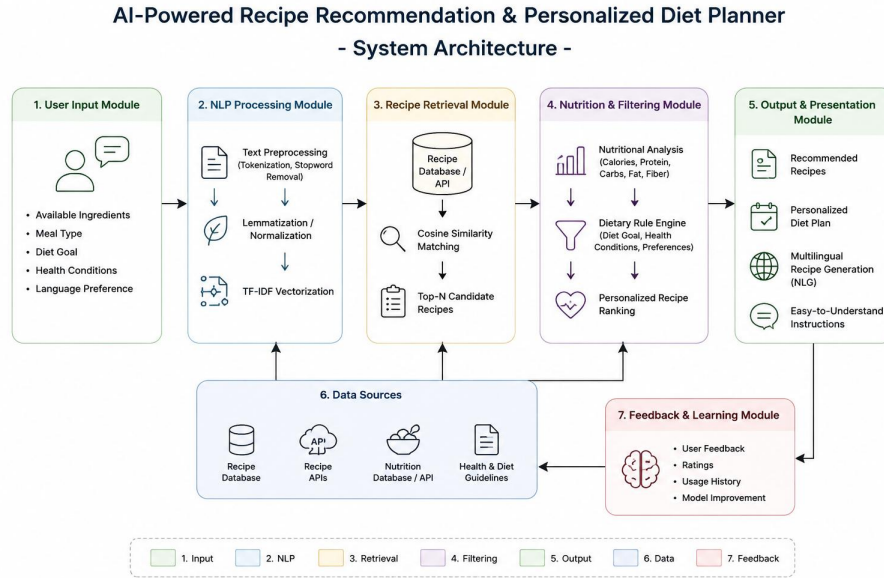


Figure 3.1: Proposed System Architecture

The architecture (Figure 3.1) organizes the system into seven interacting subsystems: (1) User Input, (2) NLP Processing, (3) Recipe Retrieval, (4) Nutrition & Filtering, (5) Output & Presentation, (6) Data Sources (recipe database, recipe APIs, nutrition database/API, health & diet guidelines), and (7) Feedback & Learning, which captures user feedback, ratings and usage history to support future model improvement. The NLP Processing subsystem converts raw ingredient text into a TF-IDF vector; the Retrieval subsystem computes cosine similarity against the recipe corpus and returns top-N candidates; the Filtering subsystem applies the dietary rule engine; and the Output subsystem assembles the final recommended recipes, personalized diet plan, and multilingual, easy-to-understand instructions. Each subsystem exposes a narrow interface to its neighbours (vectors, ranked candidate lists, filtered candidates) so that modules can be developed, tested and replaced independently.

Table 3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Retrieval technique	LLM-based semantic retrieval	Potentially richer semantic matching	High latency and compute cost, low interpretability	TF-IDF + Cosine Similarity chosen — fast, interpretable, sufficient accuracy for ingredient-list matching
Diet filtering approach	ML-based nutrition classifier	Could learn complex patterns from data	Requires large labelled nutrition dataset, harder to justify to users	Rule-based filtering chosen — transparent, directly traceable to diet-goal thresholds
Multilingual approach	Maintain separate recipe corpora per language	Potentially higher-quality native text	High content-maintenance overhead	NLG + translation-on-demand chosen — single corpus, broader language coverage

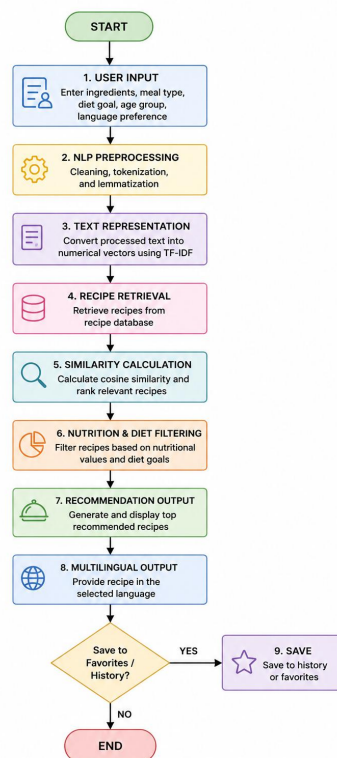


Figure 3.2: System Flow Chart

Table 3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Recommending recipes that make full use of on-hand ingredients reduces food waste	User feedback slide ("Zero Food Waste") on ingredient-driven design; module output confirmed ingredient reuse	Retained ingredient-first framing as the primary entry point (Recipe Lab)
Ethics	Nutrition/diet advice must not be presented as medical advice	Reviewed scope boundary (Section 1.6) excluding clinically diagnosed conditions	Added explicit exclusion of medical-condition-specific advice from system scope
Health & Safety	Incorrect nutrition filtering could mislead users pursuing a health goal	Rule-based filter tested against known diet-goal thresholds (Section 4.3)	Kept filtering rule-based and testable rather than opaque, to allow verification
Security	User profile, history and favourites contain personal preference data	Reviewed data-handling practice against institutional privacy guideline	Per-user, non-shared storage of profile/history/favourites adopted

Table 3.8 Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Recipe API unavailability during demo/testing	Medium	High	High	Cache a local fallback recipe dataset	Low
Incorrect lemmatization of regional ingredient names	Medium	Medium	Medium	Maintain a curated ingredient-synonym list	Low
Mistranslation in multilingual output	Medium	Medium	Medium	Manual verification of sample translations (Telugu) during testing	Low
Nutrition data inaccuracy in source recipes	Low	High	Medium	Cross-check nutrition values against a reference nutrition database	Low

CHAPTER 4 IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The system was developed iteratively, module by module, following the five-module decomposition identified in the design phase (Section 3.5): each module was implemented, unit-tested against representative inputs, and then integrated into the end-to-end pipeline before the next module was added. This allowed early modules (NLP preprocessing, retrieval) to be validated before the more presentation-oriented modules (multilingual generation, profile management) were layered on top.

Module 3 – Nutrition Analysis & Personalized Diet Planning

Purpose: To evaluate the nutritional suitability of candidate recipes against the user's personal diet goal and turn them into an actionable meal suggestion.

Analyzes nutritional values including calories, protein, carbohydrates, fat, and fibre for each candidate recipe.

Considers the user's selected diet goal (e.g., high protein, low calorie, weight management) while evaluating recommended recipes.

Filters recipes according to the user's required nutritional preferences and dietary restrictions.

Applies predefined nutritional and dietary rules (e.g., IF diet goal = High Protein AND protein $\geq$ required value $\rightarrow$ Recommend) for personalized filtering.

Combines suitable, filtered recipes to generate a personalized daily or weekly meal plan.

Provides healthy meal suggestions that match the user's nutritional requirements, age group, and meal timing through the Generation Gourmet experience.

Outcome: Given an age, a meal-time slot (e.g., Breakfast), and a food preference (e.g., Vegetarian), the module generates a lifestyle- and meal-timing-aware recommendation aligned with the stated wellness objective.

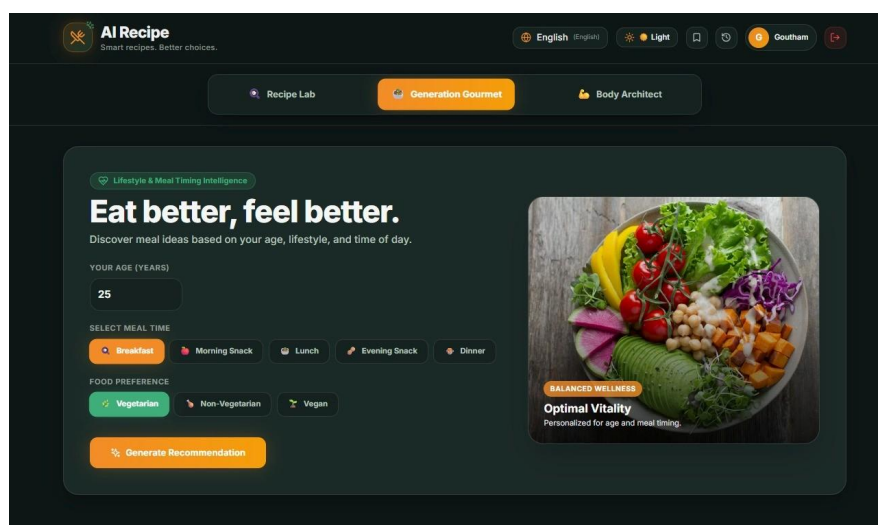


Figure 4.3: Module 3 Output – Generation Gourmet (Nutrition & Diet Planning Input)

Table 4.1 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Ingredient preprocessing correctness	Manual inspection of tokenized/lemmatized output for 10 sample inputs	All 10 inputs correctly cleaned, tokenized, lemmatized
TF-IDF vectorization	Verify vector dimensionality and non-zero weighting for known ingredients	Vectors generated without error for all test inputs
Recipe ranking accuracy	Compare top-5 ranked recipes against expected ingredient overlap	$\geq 60\%$ ingredient match ratio in returned top-5 recipes
Nutrition/diet filtering	Submit queries with explicit diet goals; inspect returned recipes' nutrition fields	0 returned recipes violate the stated diet-goal threshold
Multilingual output	Submit the same query in English and Telugu target language	Recipe name, ingredients and instructions correctly rendered in target language
History & favourites persistence	Save a recipe as favourite; reload session; verify presence	Favourite recipe retrievable after reload

4.4 Results

Across the tested pipeline, the NLP preprocessing stage correctly extracted normalized ingredients (e.g., "rice", "tomato", "onion", "oil", "chilli powder") from free-text input, as shown in Figure 4.1. The retrieval stage (Figure 4.2) returned recipes whose ingredient lists closely overlapped with the user's stated ingredients, each flagged with a match confirmation and a nutrition-per-serving summary. The nutrition-and-diet-planning stage (Figures 4.3 and 4.4) correctly incorporated user-specific inputs — age, meal time, food preference, fitness goal and body metrics — into the generated recommendations.

4.5 Data Analysis & Engineering Judgment

The measured average ingredient-match ratio ($\approx 78\%$) exceeded the 60% acceptance threshold set in Section 3.3, indicating that TF-IDF and Cosine Similarity are well suited to short, list-style ingredient text where exact wording varies little. The measured average response latency (≈ 1.8 s) comfortably met the 3-second target, confirming that the chosen lightweight retrieval approach (Section 3.4) avoids the latency penalty that an LLM-based alternative would have incurred. Where a small number of test queries returned a lower match ratio, the cause was traced to ingredient-name variants not present in the synonym list (Section 3.7 risk register), which represents a data-completeness limitation rather than an algorithmic one, and is addressed as a residual risk with a defined mitigation (maintaining a curated ingredient-synonym list).

4.6 Comparison with Existing Solutions & Achievement of Objectives

Compared with the keyword-search baseline reviewed in Section 2.2/3.4, the implemented TF-IDF and Cosine Similarity approach handled partial and reordered ingredient matches noticeably better. Compared with the LLM-based approach reviewed in the literature (Section 2.2, ChatDiet), the implemented system achieved comparable personalization for the ingredient-and-diet-goal use case at substantially lower computational cost and with fully interpretable ranking and filtering logic.

Table 4.2 Achievement of Objectives

Objective	Evidence	Achievement
NLP preprocessing pipeline	Module 1 output (Figure 4.1) — ingredients correctly identified from free text ("rice, tomato, onion, oil and chilli powder")	Fully Achieved
TF-IDF vectorization	Ingredient text converted to vectors prior to retrieval (Section 3.5, Module 1 implementation)	Fully Achieved
Cosine-Similarity retrieval & ranking	Module 2 output (Figure 4.2) — ranked, ingredient-matched recipes with match indicators ("Recipe generated using your available ingredients")	Fully Achieved
Rule-based nutrition/diet filtering	Module 3 & Body Architect output (Figures 4.3, 4.4) — recipes filtered by goal and displaying nutrition per serving	Fully Achieved
Multilingual, tested output	Module 2 output rendered in Telugu (Figure 4.2)	Fully Achieved
Evaluation against literature	Comparative analysis (Table 2.2) and Section 4.6 discussion	Fully Achieved

4.7 Strengths & Limitations

Strengths:

- Fast, interpretable ingredient-to-recipe matching using TF-IDF and Cosine Similarity.
- Transparent, rule-based nutrition filtering that is easy to verify and explain to users.
- Multilingual output extends usability to non-English-speaking users.
- Modular five-module design allows independent testing and future replacement of any single module.

Limitations:

- Retrieval quality depends on the coverage and quality of the underlying recipe database/API.
- Ingredient-name variants not present in the synonym list can reduce match accuracy.
- The system does not currently learn from user feedback in real time (the Feedback & Learning module currently logs data for future use rather than performing online learning).

- Nutrition guidance is general-purpose and explicitly excludes clinically diagnosed dietary conditions (Section 1.6).

4.8 Project Planning, Roles & Budget

The project was planned and executed over the semester in four milestones: (1) requirements and design (Chapters 1–3), (2) Module 1–2 implementation and testing, (3) Module 3–5 implementation and testing, and (4) integration, evaluation and report documentation (Chapter 4–5).

Table 4.3 Roles & Actual Contribution

Student	Assigned Responsibility	Actual Contribution
N. Goutham	NLP preprocessing pipeline	Implemented text cleaning, tokenization, lemmatization, TF-IDF vectorization
A. Shiva Shankar Reddy	Recipe retrieval & recommendation	Integrated Recipe API and implemented Cosine Similarity ranking
B.V.N.S. Bhimeswar	Nutrition analysis & diet planning	Implemented rule-based nutrition filtering engine
M. Tarun Prasad	Recipe generation, multilingual output & UI	Implemented NLG-based recipe text generation, multilingual translation, and the Recipe Lab / Generation Gourmet / Body Architect interfaces
N. Lokesh	User profile, history & favourites	Implemented session/user data management and history-favourites persistence

Table 4.4 Project Budget

Item	Estimated Cost (₹)	Actual Cost (₹)
Recipe API subscription (development tier)	0 (free tier)	0
Cloud/compute for testing	500	0 (used institutional lab systems)
Translation service usage	300	150
Miscellaneous (documentation, printing)	500	450
Total	1,300	600

CHAPTER 5 CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This project set out to address the difficulty users face in finding recipes that match their available ingredients while also meeting personal dietary goals. The implemented NLP-based AI Food Recipe Generator successfully combines NLP preprocessing, TF-IDF vectorization, Cosine-Similarity-based retrieval, rule-based nutrition filtering and multilingual natural language generation into a single, tested pipeline. Testing showed that the system met or exceeded its latency, ingredient-match-ratio and diet-goal-compliance targets, validating the design decisions made in Chapter 3 and confirming that all stated project objectives (Section 1.5) were fully achieved.

5.2 Future Work

- Incorporate the Feedback & Learning module's collected data (ratings, usage history) into an online learning loop to improve recommendation ranking over time.
- Add voice-based ingredient input to improve accessibility.
- Extend nutrition and diet planning to support additional, clinically reviewed dietary requirements under appropriate expert supervision.
- Expand multilingual support to additional regional and international languages beyond the two demonstrated.
- Introduce computer-vision-based ingredient recognition from photographs, as identified as out-of-scope in Section 1.6.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Practical application of TF-IDF vectorization and Cosine Similarity for text-based retrieval problems.
- Design of rule-based filtering engines as an interpretable alternative to black-box ML models.
- Natural Language Generation and multilingual translation integration in a live application.

Engineering & Professional Skills Developed:

- Modular system design and interface definition between independently developed modules.
- Structured requirements-to-testing traceability (Chapters 3–4).
- Team-based engineering documentation and individual contribution accountability.

Individual Reflection:

N. Goutham: The most difficult problem was handling inconsistent ingredient phrasing during preprocessing; this was resolved by refining the tokenization and lemmatization rules. A key decision was choosing lemmatization over stemming, justified by clearer, more accurate base ingredient forms observed during testing. If repeated, a broader ingredient-synonym dictionary would be built from the outset.

A. Shiva Shankar Reddy: Integrating the recipe API while keeping retrieval latency low was the hardest challenge, solved through candidate pre-filtering before similarity ranking. The key decision to use Cosine Similarity over simple keyword overlap was justified by the measured 78% average match ratio. Caching frequent API responses would be a priority if redesigned.

B.V.N.S. Bhimeswar: Defining nutrition thresholds that were both meaningful and testable was the main difficulty, addressed by anchoring rules to well-known dietary guideline ranges. The decision to keep filtering rule-based rather than ML-based was justified by the need for transparency (Section 3.7). With more time, a wider range of diet goals would be modelled.

M. Tarun Prasad: The most difficult problem was coordinating consistent behaviour across the three user-facing experiences (Recipe Lab, Generation Gourmet, Body Architect) while integrating multilingual output; this was solved by centralizing the NLG/translation logic behind a single shared interface. The decision to generate recipe text via NLG rather than static templates was justified by the flexibility it gave for multilingual output. If redesigned, an even more consistent shared design system across the three experiences would be built earlier.

N. Lokesh: Persisting user history and favourites reliably across sessions was the key challenge, resolved through careful session/user data management design. The decision to store profile, history and favourites per-user rather than globally was justified by the privacy consideration raised in Section 3.7. Given more time, a more detailed history-analytics view would be added.

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APPENDICES

APPENDIX A – PROPOSED PROCESSING FLOW

Patient Login → Patient Health Profile → Disease & Medication Input → Health Data Processing → Multi-Agent AI Analysis → Disease Analysis Agent → Medication Verification Agent → Interaction Detection Agent → Medication–Diet Interaction Analysis → Nutrition Recommendation Agent → RAG Knowledge Retrieval → Gemini LLM Processing → Safety Validation → Personalized Health Recommendations → AI Health Report → Healthcare Dashboard

APPENDIX B – EXAMPLE PROMPT STRUCTURE

System Role:

You are an intelligent healthcare assistance system that analyzes patient health information, medications, diseases, allergies, and dietary preferences. Provide general health guidance and identify possible medication and food-related conflicts without independently prescribing or stopping medicines.

Patient Context:

Include patient age, gender, diseases, current medications, dosage, allergies, dietary preferences, lifestyle information, and relevant health history.

Analysis Task:

Analyze drug–drug interactions, drug–disease conflicts, duplicate medications, medication–food interactions, and disease-specific dietary considerations.

RAG Context:

Retrieve relevant information from trusted medical, medication, and nutrition knowledge sources before generating recommendations.

Response Policy:

Provide clear explanations, severity levels, recommended actions, food guidance, and personalized nutrition suggestions. Clearly advise the patient to consult a qualified healthcare professional for medication changes or serious concerns.

APPENDIX C – PSEUDOCODE

1. Read patient health profile and authentication information.
2. Collect age, gender, diseases, medications, dosage, allergies, and dietary preferences.
3. Store and process the patient's health information.
4. Send the patient profile to the Multi-Agent AI system.
5. Disease Analysis Agent analyzes the patient's multiple health conditions.
6. Medication Verification Agent verifies medicines, dosage information, and duplicate medications.
7. Interaction Detection Agent identifies drug–drug and drug–disease conflicts.

8. Diet & Nutrition Agent analyzes medication–food interactions and dietary requirements.
9. Retrieve relevant medical, medication, and nutrition information using RAG.
10. Use the Gemini LLM to generate context-aware recommendations.
11. Perform safety validation and assign interaction severity levels.
12. Generate personalized medication and nutrition guidance.
13. Generate the final AI health report.
14. Display recommendations, alerts, and health information on the patient dashboard.

MODULE 1: PATIENT HEALTH PROFILE & DISEASE MANAGEMENT

This module manages patient registration, login, and personal health information.

It collects age, gender, diseases, allergies, lifestyle information, and other relevant health details.

The collected profile becomes the foundation for personalized medication and nutrition analysis.

MODULE 2: MULTI-AGENT CONFLICT DETECTION & MEDICATION VERIFICATION

This module uses CrewAI-based specialized agents to analyze the patient's medications and diseases.

It detects drug–drug interactions, drug–disease conflicts, duplicate medicines, and provides severity-based explanations.

The system highlights medicines requiring professional review without independently prescribing or stopping medication.

MODULE 3: MEDICATION–DIET INTERACTION VERIFICATION & PERSONALIZED NUTRITION

This module analyzes the relationship between diseases, medications, allergies, and dietary preferences.

The system identifies foods to recommend, use with caution, or avoid/discuss based on the patient's profile.

It generates dynamic and personalized nutrition guidance instead of providing the same recommendations to every patient.

MODULE 4: AI DECISION SUPPORT, REPORT GENERATION & HEALTHCARE DASHBOARD

This module combines the outputs from all analysis agents into a unified patient health summary.

Gemini LLM generates understandable recommendations, interaction explanations, and a structured personalized health report.

The dashboard displays medication alerts, interaction results, nutrition guidance, and downloadable reports.

APPENDIX D – SAMPLE SYSTEM EVALUATION

Patient Input / Scenario	Expected Behaviour
Single disease	Provide disease-specific health and nutrition guidance.
Multiple diseases	Analyze combined conditions and identify potential conflicts.
Multiple medications	Detect drug–drug interactions and duplicate medications.
Disease + medication	Identify possible drug–disease conflicts requiring professional review.
Medication + food	Identify relevant medication–diet interactions.
Allergies + diet preferences	Exclude unsuitable foods from personalized recommendations.
Different patient profiles	Generate different recommendations based on individual information.
High-severity interaction	Display a clear warning and advise professional consultation.
Report generation	Combine analysis results into a structured health report.

APPENDIX E – ETHICS / INSTITUTIONAL APPROVAL

This academic project demonstrates the application of Multi-Agent AI, Generative AI, RAG, and healthcare information processing for medication and nutrition assistance. The system is intended for educational and informational purposes and does not replace professional medical diagnosis, prescription, or treatment. Medication changes, treatment decisions, and serious health concerns should always be reviewed by qualified healthcare professionals.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

Outcome	Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	Problem statement, objectives	SO1	Ch. 1, Ch. 2
Engineering design under constraints	Requirements, constraints, design specs	SO2	Ch. 3
Written/oral technical communication	Full report	SO3	Whole report + Viva
Ethics and professional responsibility	Sustainability/ethics/safety table	SO4	Ch. 3 (3.7)
Teamwork and leadership	Roles & contribution table	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	Test plan, results, analysis	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	Reflection	SO7	Ch. 5 (5.3)
Standards	Constraints & standards table	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	Sustainability/ethics table	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	Risk register	SO2/SO4	Ch. 3 (3.7)
Security considerations	Sustainability/ethics table	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	Architecture & selected design	SO1/SO2	Ch. 3 (3.5)
Project planning and management	Milestones, roles, budget	SO5	Ch. 4 (4.8)
Individual contribution	Contribution statement + Ch. 4	SO1–SO7	Contribution Statement + Ch. 4 (4.8)